

H524 Introduction to Biostatistics

Week 1 Lab: Introduction to R and Data Visualization

Fall 2025

Lab Duration: 110 minutes (8:00am – 9:50am)

Format: Online

Tools Required: R, RStudio

1 Learning Objectives

By the end of this lab, you will be able to:

1. Set up R and RStudio for biostatistical analysis
2. Create and manipulate basic R data structures
3. Generate descriptive statistics for health data
4. Create publication-quality visualizations for different data types

2 Lab Setup

2.1 Required Software

1. **R:** Download from <https://cran.r-project.org/>
2. **RStudio:** Download from <https://posit.co/products/open-source/rstudio/>

2.2 Verify Installation

Open RStudio and run the following code to verify your setup:

```
# Check R version
R.version.string

# Check working directory
getwd()

# Test basic functionality
x <- c(1, 2, 3, 4, 5)
mean(x)
```

3 Part 1: R Fundamentals (30 minutes)

3.1 Data Types and Structures

Exercise 1.1: Create variables of different data types relevant to health research.

Note: The health data used in these exercises are simulated values designed for demonstration purposes. They represent typical ranges found in clinical practice but are not from actual patient records.

```

# Numeric data (simulated systolic/diastolic blood pressure values)
systolic_bp <- c(120, 135, 142, 118, 160, 125, 138)
diastolic_bp <- c(80, 85, 92, 75, 95, 82, 88)

# Character data
patient_id <- c("P001", "P002", "P003", "P004", "P005", "P006", "P007")
gender <- c("Male", "Female", "Female", "Male", "Male", "Female", "Male")

# Factor data (categorical)
treatment_group <- factor(c("Control", "Treatment", "Treatment",
                           "Control", "Treatment", "Control", "Treatment"))

# Logical data
hypertensive <- systolic_bp >= 140 | diastolic_bp >= 90

# Display the data types
class(systolic_bp)
class(gender)
class(treatment_group)
class(hypertensive)

```

Exercise 1.2: Create a data frame combining health variables.

```

# Create a comprehensive health dataset (based on 2024 materials)
health_data <- data.frame(
  patient_id = patient_id,
  gender = gender,
  systolic_bp = systolic_bp,
  diastolic_bp = diastolic_bp,
  treatment_group = treatment_group,
  hypertensive = hypertensive,
  age = c(45, 52, 38, 61, 47, 34, 55),
  weight_kg = c(75, 68, 82, 71, 88, 63, 79),
  height_cm = c(170, 165, 178, 172, 180, 160, 175)
)

# View the data structure
str(health_data)
head(health_data)
summary(health_data)

```

4 Part 2: Descriptive Statistics (25 minutes)

Exercise 2.1: Calculate measures of central tendency and variability.

```

# Measures of central tendency
mean_systolic <- mean(health_data$systolic_bp)
median_systolic <- median(health_data$systolic_bp)

# Measures of variability
sd_systolic <- sd(health_data$systolic_bp)

```

```

var_systolic <- var(health_data$systolic_bp)
range_systolic <- range(health_data$systolic_bp)
iqr_systolic <- IQR(health_data$systolic_bp)

# Display results
cat("Mean systolic BP:", round(mean_systolic, 2), "mmHg\n")
cat("Median systolic BP:", median_systolic, "mmHg\n")
cat("Standard deviation:", round(sd_systolic, 2), "mmHg\n")
cat("Range:", range_systolic[1], "-", range_systolic[2], "mmHg\n")

```

Exercise 2.2: Generate descriptive statistics by group.

```

# Statistics by gender
aggregate(systolic_bp ~ gender, data = health_data, FUN = mean)
aggregate(systolic_bp ~ gender, data = health_data, FUN = sd)

# Statistics by treatment group
aggregate(systolic_bp ~ treatment_group, data = health_data, FUN = summary)

# Frequency tables for categorical data
table(health_data$gender)
table(health_data$treatment_group)
prop.table(table(health_data$hypertensive))

```

5 Part 3: Data Visualization (35 minutes)

5.1 Geographic Health Mapping

Exercise 3.1: Create health-focused maps (mirroring 2024 approach).

```

# Install required packages for mapping (if not already installed)
# install.packages(c("maps", "mapdata", "ggplot2"))
library(maps)
library(mapdata)
library(ggplot2)

# Create simulated state-level health data
state_health <- data.frame(
  state = c("oregon", "washington", "california", "nevada", "idaho"),
  diabetes_rate = c(8.2, 9.1, 9.7, 10.8, 8.9),
  obesity_rate = c(28.7, 27.9, 25.1, 26.8, 29.2)
)

# Get US state map data
states_map <- map_data("state")

# Merge health data with map coordinates
map_data <- merge(states_map, state_health, by.x = "region", by.y = "state")

# Create diabetes prevalence map
ggplot(map_data, aes(x = long, y = lat, group = group, fill = diabetes_rate)) +
  geom_polygon(color = "white") +

```

```

scale_fill_gradient(low = "lightblue", high = "darkred",
                    name = "Diabetes Rate (%)") +
coord_quickmap() +
theme_void() +
labs(title = "Diabetes Prevalence by State",
     subtitle = "Simulated data for demonstration")

```

Output: This code will generate a choropleth map showing diabetes prevalence rates across Pacific Northwest states. Higher rates appear in darker red, with Washington and Nevada showing elevated diabetes prevalence compared to Oregon and California.

Figure 1: Geographic visualization of diabetes prevalence rates (code output shown)

5.2 Visualizations for Continuous Data

Exercise 3.2: Create histograms and box plots with enhanced features.

```

# Histogram of systolic blood pressure
hist(health_data$systolic_bp,
     main = "Distribution of Systolic Blood Pressure",
     xlab = "Systolic BP (mmHg)",
     ylab = "Frequency",
     col = "lightblue",
     border = "white",
     breaks = 5)

# Add vertical line for mean
abline(v = mean_systolic, col = "red", lwd = 2, lty = 2)
legend("topright", "Mean", col = "red", lwd = 2, lty = 2)

# Box plot by gender
boxplot(systolic_bp ~ gender,
       data = health_data,
       main = "Systolic BP by Gender",
       ylab = "Systolic BP (mmHg)",
       col = c("pink", "lightblue"))

```

Output: Histogram displays systolic blood pressure distribution with blue bars, white borders, and a red dashed vertical line indicating the mean value (approximately 135 mmHg).

Figure 2: Example histogram showing blood pressure distribution with mean line

Output: Side-by-side box plots comparing systolic blood pressure between males (light blue) and females (pink), showing median, quartiles, and any outliers.

Figure 3: Example box plot comparing blood pressure by gender

5.3 Advanced Health Visualizations

Exercise 3.3: Create BMI analysis with multiple plot types.

```
# Calculate BMI and categorize
health_data$bmi <- health_data$weight_kg / (health_data$height_cm/100)^2
health_data$bmi_category <- cut(health_data$bmi,
                               breaks = c(0, 18.5, 25, 30, Inf),
                               labels = c("Underweight", "Normal", "Overweight", "Obese"))

# Create comprehensive BMI visualization
par(mfrow = c(2, 2)) # 2x2 plot layout

# 1. BMI histogram with normal curve overlay
hist(health_data$bmi, probability = TRUE,
     main = "BMI Distribution", xlab = "BMI", col = "lightblue")
curve(dnorm(x, mean = mean(health_data$bmi), sd = sd(health_data$bmi)),
     add = TRUE, col = "red", lwd = 2)

# 2. BMI by gender boxplot
boxplot(bmi ~ gender, data = health_data,
        main = "BMI by Gender", ylab = "BMI", col = c("pink", "lightblue"))

# 3. BMI categories bar chart
bmi_table <- table(health_data$bmi_category)
barplot(bmi_table, main = "BMI Categories",
        col = c("lightgreen", "yellow", "orange", "red"),
        ylab = "Frequency")

# 4. Scatter plot: BMI vs Systolic BP
plot(health_data$bmi, health_data$systolic_bp,
     pch = 19, col = as.numeric(health_data$gender),
     main = "BMI vs Systolic BP",
     xlab = "BMI", ylab = "Systolic BP (mmHg)")
legend("topright", levels(health_data$gender),
     col = 1:2, pch = 19)
abline(lm(systolic_bp ~ bmi, data = health_data), col = "red", lwd = 2)
```

Output: Four-panel visualization showing: (1) BMI distribution histogram with normal curve overlay, (2) BMI comparison by gender using box plots, (3) BMI category frequencies in a bar chart, and (4) scatter plot of BMI vs systolic blood pressure with regression line and gender-coded points.

Figure 4: Comprehensive BMI analysis: distribution, gender comparison, categories, and correlation with blood pressure

5.4 Visualizations for Categorical Data

Exercise 3.4: Create enhanced bar charts and pie charts.

```
# Bar chart for treatment groups
treatment_counts <- table(health_data$treatment_group)
```

```

barplot(treatment_counts,
       main = "Distribution of Treatment Groups",
       ylab = "Frequency",
       col = c("#FF6B35", "#F7931E", "#FFAA1D"))

# Pie chart for gender distribution
gender_counts <- table(health_data$gender)
pie(gender_counts,
    main = "Gender Distribution",
    col = c("pink", "lightblue"))

```

Output: Vertical bar chart showing treatment group frequencies with orange-colored bars and clear group labels.

Figure 5: Example bar chart for categorical health data

Output: Pie chart displaying gender distribution with pink slice for females and light blue slice for males, including percentage labels.

Figure 6: Example pie chart showing proportions

5.5 Advanced Visualizations

Exercise 3.3: Create scatter plots and add trend lines.

```

# Scatter plot of systolic vs diastolic BP
plot(health_data$diastolic_bp, health_data$systolic_bp,
     xlab = "Diastolic BP (mmHg)",
     ylab = "Systolic BP (mmHg)",
     main = "Systolic vs Diastolic Blood Pressure",
     pch = 19,
     col = as.numeric(health_data$gender))

# Add legend
legend("topleft", levels(health_data$gender),
      col = 1:2, pch = 19)

# Add regression line
abline(lm(systolic_bp ~ diastolic_bp, data = health_data),
      col = "red", lwd = 2)

```

Output: Scatter plot displays diastolic BP (x-axis) vs systolic BP (y-axis) with color-coded points by gender and a red regression line showing positive correlation.

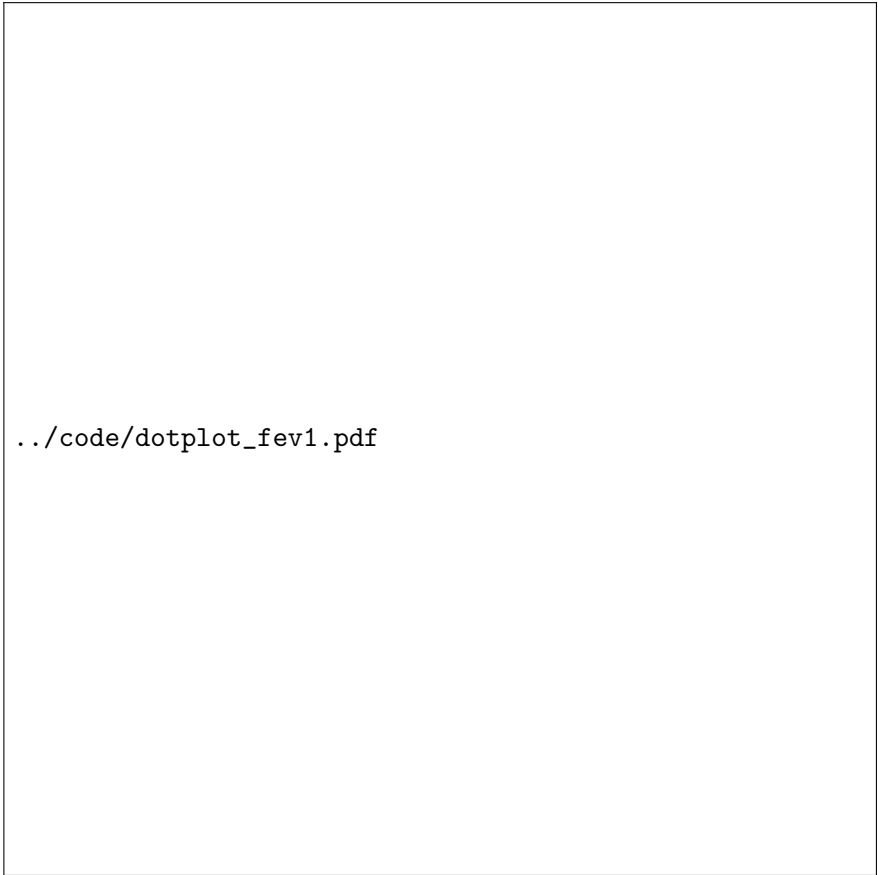
Figure 7: Example scatter plot showing relationship between blood pressure variables

6 Lab Assignment

6.1 Data Analysis Task

Use the FEV1 lung function data from the course materials:

```
# FEV1 data from Pagano & Gauvreau study
fev1_data <- data.frame(
  subject_id = 1:13,
  gender = c("F", "F", "F", "F", "F", "F", "M", "M", "M", "M", "M", "M", "M"),
  fev1_liters = c(2.46, 2.65, 2.69, 2.76, 2.85, 3.15, 2.47, 2.77, 2.78,
                  2.81, 3.09, 3.18, 3.46)
)
```



../code/dotplot_fev1.pdf

Figure 8: Example dot plot showing individual FEV1 measurements

Assignment Tasks:

1. Calculate descriptive statistics (mean, median, SD, range) for FEV1 by gender
2. Create comprehensive visualizations including:
 - Geographic map showing state-level respiratory disease rates
 - Multi-panel plot combining histogram, box plot, and dot plot

- BMI-FEV1 correlation analysis with trend lines
 - Risk factor visualization (smoking, age, gender effects)
3. Apply advanced analysis approaches:
- Risk ratio calculations where appropriate
 - Correlation testing between variables
 - Age-adjustment techniques for rate comparisons
 - Professional publication-ready plot formatting

6.2 Deliverables

Submit the following files via Canvas:

1. `week1_lab_[lastname].R` - R script with all exercises
2. `week1_lab_report_[lastname].pdf` - Brief report (1-2 pages) including:
 - Summary statistics tables
 - Key visualizations
 - Brief interpretation of FEV1 findings

7 Additional Resources

- **R Documentation:** <https://www.rdocumentation.org/>
- **RStudio Cheatsheets:** <https://posit.co/resources/cheatsheets/>
- **Biostatistics with R:** <https://cran.r-project.org/web/views/MedicalImaging.html>